

# Unit 19: Electronic Devices & Circuits

Week 1 Deliverables Form (PoE A01) — Microsoft Forms Import Template

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## ■ Form Configuration Settings (For Form Owner):

- **Who can fill out this form:** Only people in my organization can respond
  - **Record name:** Checked (Auto-captures student email & full name via Microsoft 365)
  - **One response per person:** Checked (Prevents duplicate entries)
  - **Allow response editing after submission:** Checked (Enables iterative updates throughout the 160-minute practical)
  - **Send email receipt to respondents:** Checked (Gives students a verifiable record of their engineering evidence)
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## Section 1: Engineering Role & Circuit Identification

*Workplace Context: You are joining the prototype lab as a Junior Electronics Technician. A low-voltage LED indicator board has failed production testing, and your supervisor needs you to investigate and verify it.*

### 1. Student Full Name

*Type: Text (Single line)*

*Required: Yes*

### 2. Student ID / College Number

*Type: Text (Single line)*

*Required: Yes*

### 3. Engineering Requirement: What was the circuit designed to do when operating normally?

*Type: Text (Long answer)*

*Prompt: Describe the expected operation when the switch is closed and power is supplied.*

*Required: Yes*

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## Section 2: Safe Electronic Working Practices & Emergency Procedures

*Safe laboratory standard operating procedures, hazard identification, and isolation.*

### 4. Pre-Power Verification Checklist

*Type: Choice (Multiple answers allowed)*

*Prompt: Confirm which mandatory pre-power checks you completed before connecting power:*

- ☐ Work bench clean, dry, and clear of loose wire clippings or uninsulated metal
- ☐ DC Bench Power Supply switched OFF and verified at 5.0 V with current limit  $\leq 100$  mA
- ☐ LED polarity verified (Anode to +5V rail, Cathode to 0V ground)
- ☐ Current-limiting resistor value checked ( $220\ \Omega$  /  $330\ \Omega$ ) - NOT a  $0\ \Omega$  short link
- ☐ Breadboard rows inspected to confirm components share valid terminal columns

*Required: Yes*

### 5. Emergency Scenario: Smoke or Overheating

*Type: Choice*

*Prompt: If smoke or excessive heat is detected from a breadboard during testing, what is the mandatory FIRST action?*

- [ ] Touch the components to identify which one is hot
- [ ] Disconnect / isolate the electrical power supply immediately
- [ ] Continue measuring with the multimeter to record the fault voltage
- [ ] Swap the LED component immediately

**Required:** Yes (Correct: Disconnect/isolate power)

## 6. Safety Decision: Incorrect Supply Voltage Setting

**Type:** Text (Long answer)

**Prompt:** A 5 V circuit is connected to a PSU set to 12 V. Your partner says "It's only for a few seconds, it will be fine." Explain what you should do and why.

**Required:** Yes

## 7. Laboratory Emergency Stop Location

**Type:** Text (Single line)

**Prompt:** State the location of the main electrical isolation / emergency stop button in your laboratory.

**Required:** Yes

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## Section 3: Circuit Schematic Interpretation & Virtual Simulation

Schematic reference: +5V Supply -> S1 (Switch) -> R1 (330Ω/220Ω) -> D1 (LED) -> 0V Return.

## 8. Component Function Matching

**Type:** Text (Single line)

**Prompt:** State the function of R1 and D1.

**Required:** Yes

## 9. Current Path Order

**Type:** Text (Single line)

**Prompt:** Starting from +5 V, list the exact order current passes through the circuit to 0V.

**Required:** Yes (Expected: +5V -> Switch S1 -> Resistor R1 -> LED D1 -> 0V)

## 10. What could happen to the LED if R1 was accidentally replaced with a 0 Ω wire link?

**Type:** Text (Long answer)

**Required:** Yes

## 11. Simulation Verification: What was your simulated baseline current and LED forward voltage?

**Type:** Text (Single line)

**Prompt:** Record your simulated circuit current ( $I_{total}$ ) and LED forward voltage ( $V_{LED}$ ) from Multisim / Tinkercad.

**Required:** Yes

## 12. Upload a screenshot of your completed circuit simulation in Multisim / Tinkercad

**Type:** File Upload (Image)

**File Limit:** 1

**Max Size:** 10MB

**Required:** Yes

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## Section 4: Systematic Troubleshooting & Diagnostic Measurements

Diagnostic cycle: Observe -> Predict -> Test -> Locate -> Fix -> Verify.

**13. Initial Observation: What was the observed symptom on the faulty board?**

*Type: Text (Long answer)*

*Required: Yes*

**14. Hypothesis / Initial Prediction: What did you predict was causing the fault before changing anything?**

*Type: Text (Long answer)*

*Required: Yes*

**15. Diagnostic Test Measurements (Paste from Website Measurement Logger)**

*Type: Text (Long answer)*

*Prompt: Paste your formatted measurement summary generated by the "Copy Summary for Microsoft Form" tool on the website (including Vs, VR1, V\_LED, and I\_total).*

*Required: Yes*

**16. Fault Root Cause: What was the confirmed fault?**

*Type: Choice*

- ☐ Fault A: LED installed backwards (Reverse-biased open circuit)
- ☐ Fault B: Open resistor connection / missing jumper wire
- ☐ Fault C: Incorrect breadboard terminal row misalignment
- ☐ Fault D: Missing 0V supply ground return rail
- ☐ Fault E: Incorrect high resistor value (e.g. 220 k $\Omega$  instead of 220  $\Omega$ )
- ☐ Fault F: Loose / high-resistance jumper wire connection
- ☐ Stretch Fault: Secondary / intermittent wiring fault

*Required: Yes*

**17. Corrective Repair Action: What physical change did you make to fix the circuit?**

*Type: Text (Long answer)*

*Required: Yes*

**18. Verification: What measurements and visual proof confirm that the repair was successful?**

*Type: Text (Long answer)*

*Prompt: State the final verified LED state, measured voltage VR1, and circuit current I\_total.*

*Required: Yes*

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## Section 5: Physical Circuit Photographic Evidence

**19. Upload a clear photograph of your verified, operational physical breadboard circuit**

*Type: File Upload (Image)*

*File Limit: 1*

*Max Size: 10MB*

*Required: Yes*

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## Section 6: Engineer Exit Ticket & Professional Sign-Off

**20. Exit Ticket Q1: Why is the statement "I think the resistor is faulty" not enough to justify replacing it?**

*Type: Text (Long answer)*

*Required: Yes*

**21. Exit Ticket Q2: If the PSU is 5.0 V but you measure 0.0 V across the resistor, give TWO possible causes.**

*Type: Text (Long answer)*

*Required: Yes*

**22. Technician Sign-Off Decision: Would you sign off this prototype for production integration?**

*Type: Choice*

☐ YES - Fully verified against specifications

☐ YES, WITH CONDITIONS - Minor wire routing rework needed

☐ NO - Requires further bench testing

*Required: Yes*

**23. What is one troubleshooting skill you want to improve in next week's Diode Lab?**

*Type: Text (Single line)*

*Required: Yes*